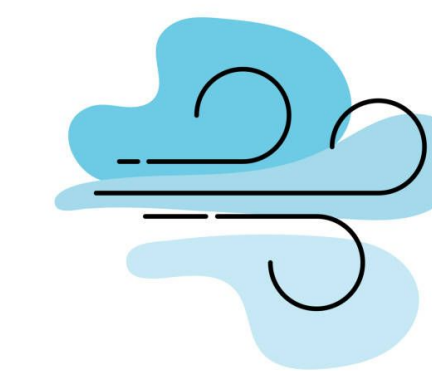




UTD-011 Machine Learning-Enabled Condition Monitoring of Wind Turbines Using High-Resolution Electrical Signals



Team Members: Amanda Yin, Christopher Huang, Emily Yin, Irene Liu, Tianyu Lu
Advisor: Jie Zhang **Graduate Assistants:** Jingyi Yan, Fazlur Rahman Bin Karim

Problem Statement and Motivation

- Wind turbines are critical components of renewable energy infrastructure. Out of total wind turbine expenditure, the cost of maintaining wind turbines is 38%, and the generator is responsible for 17% of all wind turbine failure.
- The main goal is to provide a cost-effective and reliable AI- powered condition monitoring system (CMS) for wind turbines that maximizes operational lifetime, reduces downtime, and enhances overall efficiency. This would contribute to making wind energy more competitive and sustainable in the energy market.

State-of-The-Art

Industry Standard Condition Monitoring System (CMS)

Key Components Monitored

- Gearbox: Vibration, oil temperature, particle count.
- Generator: Temperature, vibration, insulation resistance, SCADA data.
- Rotor Blades: Acoustic emissions, strain gauges.
- Main Shaft and Bearings: Vibration and temperature sensors.
- Tower & Structure: Accelerometers and tilt sensors.

Types of Sensors Used

- Accelerometers (vibration)
- Thermocouples (temperature)
- Oil debris sensors
- Current and voltage transducers

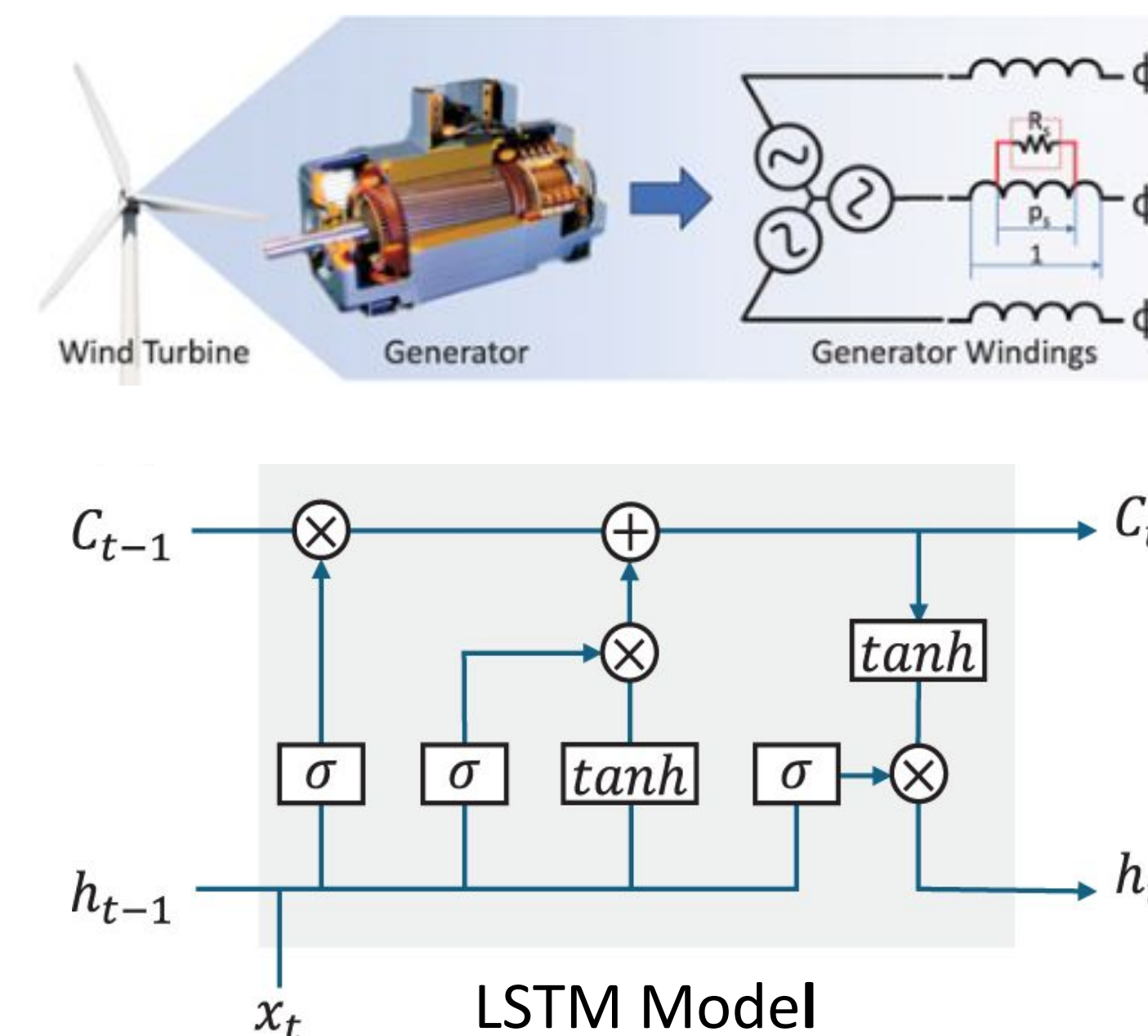
Project Overview and Technical Approach

Using a Recurrent Neural Network (RNN) to Analyze Wind Turbine Data

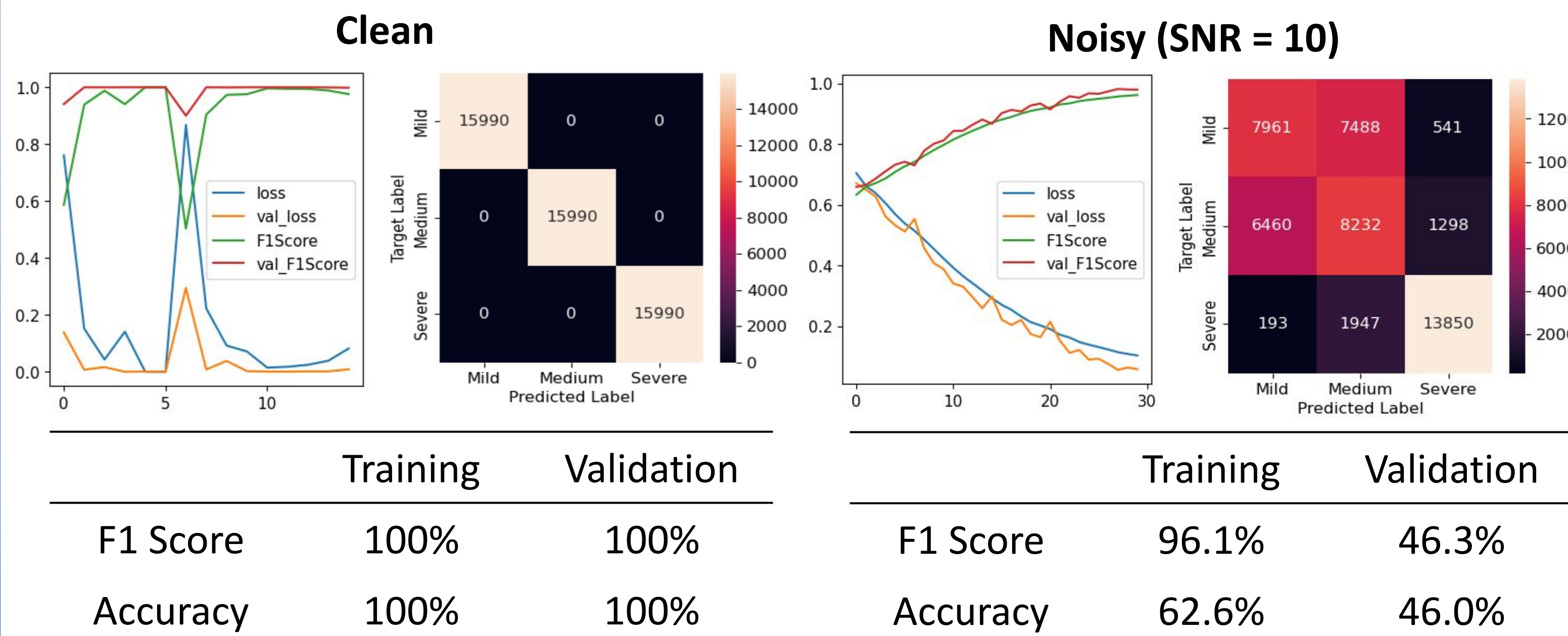
- The project consisted of learning the basic components of wind turbines, which was followed by learning data visualization and machine learning in Python.
- The CMS is a machine learning model (RNN) that was trained on datasets that contained wind turbine data in order to detect faults.
- The CMS is used to monitor the data within the generator of the wind turbine, more specifically the generator windings.

Technical Approach

- For our CMS, an RNN was used, specifically a Long Short-Term Memory (LSTM) model, in order to overcome the vanishing/exploding gradient problem of vanilla RNNs. RNNs are designed to handle sequential (times series) data with internal memory stored in "hidden states", allowing information to persist across time steps.



Results



Conclusions and Outlook

- Data-driven CMS' utilizing Machine Learning models are effective at detecting faults in wind turbines and can be used reliably for system maintenance.
- As the market for wind energy grows, so will the market for wind turbine operations and maintenance, as shown in the graph below. This project offers a potential AI and data-powered solution to a critical need that will only grow in importance as the industry expands in the future.

WIND TURBINE OPERATIONS AND MAINTENANCE MARKET SIZE, 2020 TO 2030 (USD BILLION)



References

Yan, J., Senemmar, S., and Zhang, J. (December 9, 2024). "Bi-Level Interturn Short-Circuit Fault Monitoring for Wind Turbine Generators With Benchmark Dataset Development." *ASME. J. Mech. Des.* April 2025; 147(4): 041704.
<https://doi.org/10.1115/1.4067056>